Targeting Scar States Using Polynomial Filtering in the PXP Hamiltonian

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July 3, 2026

Eigenstate Thermalization Hypothesis (ETH)

- Explains thermalization in isolated quantum many-body systems.
- Local observables relax to match expected values from statistical mechanics.
- Individual generic eigenstates mimic thermal ensembles.

Important: ETH Violations

Some systems have rare, unusual eigenstates that strongly violate ETH and fail to thermalize.

The PXP Model & Quantum Many-Body Scars

- Models projected spin-flip dynamics on a 1D Rydberg atom chain:

$$H = \sum_{i=0}^{L-1} P_i X_{(i+1)\%L} P_{(i+2)\%L} \tag{1}$$

  where $P_i = \frac{1}{2}(\mathbb{1} - Z_i)$ projects site i onto its ground state $|0\rangle$.
- ETH-violating eigenstates form **quantum many-body scar states**.
- Scars are rare, evenly spaced, and possess unusually low entanglement.
- But there is the usual problem, the Hilbert space grows exponentially.

Problem of the Project

Can we use Chebyshev polynomial filtering to target the scar states without doing full exact diagonalization?

Understanding the PXP Hamiltonian

Adjacency graph etc.